QT

MGU BCA - 3RD SEM

# Quantitative Techniques

Complete Important Questions

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MODULE 1

# Introduction to Statistics

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**EXAM PATTERN (from real MGU question papers)**

EXAM PATTERN (QT): The real MG3CCRBCA200 paper pattern is UNKNOWN - no Honours model paper and no past BCA paper exists online (first exam 22-10-2025 was never published). Closest evidence: MGU B.Com 'Quantitative Techniques for Business' papers (same university, overlapping statistics syllabus) run 80 marks / 3 hours: Part A answer any 10 x 2 marks = 20 (short theory + one-line numericals), Part B answer any 6 x 5 marks = 30 (medium theory + numericals), Part C answer any 2 x 15 marks = 30 (long numericals: combined mean/SD/CV two-firm problem repeated 3x, Bayes theorem). QT is a 4-credit course, so the Honours paper may instead follow a 70-mark / 2-hour format (like DBMS) - this is NOT confirmed. Prepare every module: Part A 2-mark questions span all modules.

Every module, every important, repeated and sure-mark question for Quantitative Techniques, each with a full-mark model answer.

## 1 Define statistics and explain its meaning.

**IMPORTANT**

Statistics is the science of collecting, organising, analysing, interpreting and presenting numerical data so that sensible conclusions can be drawn from them.

The word statistics comes from the Latin word 'status' meaning a state, because rulers first collected facts about population, army and revenue of the state.

In simple meaning, statistics is the set of methods used to handle numerical facts. For example, the marks of 500 students in a college are just raw numbers, but when we find the average mark and the highest and lowest marks, we are using statistics.

Statistics does not deal with one or two isolated facts. It always deals with a large collection of facts, because conclusions are drawn from the group as a whole.

The main purpose of statistics is to turn confusing raw data into clear information. It replaces guesswork with a systematic way of studying facts and making decisions.

For example, a shopkeeper who records daily sales for a month and finds which item sold most is using statistics in the simplest form.

In short, statistics is the science of learning from numbers.

## 2 Explain the two senses in which the word statistics is used.

IMPORTANT

The word statistics is used in two different senses: singular and plural.

In the singular sense, statistics means the science of statistics. It is the body of methods used to collect, analyse and interpret numerical data, just like physics or mathematics is a science.

In the plural sense, statistics means statistical data. Here it refers to the numerical facts themselves, like the population of India, the rainfall in Kerala, or the prices of rice in a market.

For example, the statement 'the statistics of road accidents show a rise' uses statistics in the plural sense, meaning the actual figures.

On the other hand, the statement 'she studied statistics in college' uses statistics in the singular sense, meaning the subject or science.

Both senses are connected: the science of statistics (singular) is applied to statistical data (plural) to draw useful conclusions.

## 3 'Statistics is the science of averages.' Explain.

IMPORTANT

The statement 'statistics is the science of averages' means that an average is the most common and most useful tool in statistics.

An average represents the whole group of numbers by one single figure. For example, instead of remembering the salaries of 200 workers, one average salary gives the general idea of the group.

Averages make large data easy to understand. The average marks of a class tell us quickly whether the class performed well or badly.

Averages also help in comparison. We can compare the average rainfall of two districts, or the average profit of a company in two years, using a single figure.

Economists and businessmen use averages daily, for example average price, average cost, average production and average demand.

However, the statement is not fully complete. Averages are very important, but statistics also uses other tools like diagrams, correlation and forecasting, so statistics is much more than just averages.

In short, statistics can be called the science of averages because the average is its most widely used product, though not its only one.

## 4 Explain the functions of statistics. (2024, 2022)

REPEATED

One function of statistics is to simplify complex data. Raw data like the monthly sales of 100 shops is too large to understand, but an average or a percentage makes it simple.

Second, statistics helps in comparison. Figures from different years, places or companies can be compared only after they are put into a common form like averages or percentages.

Third, statistics helps in forming policies. A government studies unemployment figures before launching a job scheme, and a company studies sales data before expanding production.

Fourth, statistics helps in forecasting. Past data studied with statistical methods can predict future trends, for example predicting next year's demand from the sales of past five years.

Fifth, statistics helps in measuring relationships. It can show whether two things move together, like whether advertising expenditure and sales are related.

Sixth, statistics helps in testing old ideas. An old belief, like 'demand falls when price rises', can be checked with real data instead of mere opinion.

Finally, statistics enlarges human knowledge. Scientific research in medicine, agriculture and social sciences depends on statistical analysis of experiments.

In short, statistics collects, simplifies, compares and interprets data so that sound decisions and predictions become possible.

## 5 Explain the uses of statistics in business and economics.

IMPORTANT

In business, statistics is used to plan production. A company studies past sales figures to decide how much to produce next month, so that there is neither shortage nor waste.

Statistics helps in studying the market. Before launching a new product, firms collect data on customer preferences, income and buying habits to judge the demand.

It is used in quality control. A factory checks a small sample of its products to judge whether the whole production is of good quality, instead of checking every single piece.

It helps in fixing prices and measuring profit. Costs, sales and profits of different periods or branches are compared using averages and percentages.

In economics, statistics is used to measure the health of the economy. Figures like national income, price index, unemployment rate and growth rate are all statistical products.

It helps in studying trade and prices. A businessman watches the price index and market trends to decide when to buy raw materials and when to sell finished goods.

Governments use statistics for economic planning. The five-year plans and budgets of India are

prepared from data on population, agriculture, industry and trade.

In short, statistics is the backbone of business decisions and economic policy, because both run on facts, not on guesses.

## 6 'Statistics can be misused.' Explain with examples. (2018, 2019)

REPEATED

Statistics can be misused means that the same set of figures can be presented in a dishonest way to create a false impression.

One common misuse is choosing a convenient time period. A company that grew from Rs. 10 lakh profit to Rs. 11 lakh in ten years can still say 'profit doubled' if it picks a low-profit base year.

A second misuse is the wrong type of average. If five workers earn Rs. 8,000 each and the owner earns Rs. 5,00,000, the company can proudly say 'average salary Rs. 90,000', which hides the truth.

A third misuse is misleading diagrams. A graph with a very small scale can make a tiny rise in sales look like a huge jump, fooling the viewer at first glance.

A fourth misuse is biased sampling. A company can test only its satisfied customers and then claim '90 percent of customers are happy', while the unhappy ones were never asked.

Another misuse is quoting percentages without the base. A shop can claim 'sales increased 100 percent' when sales went from 2 items to 4 items, which sounds big but is actually small.

The misuse is not the fault of statistics itself. It is the fault of dishonest users, and the remedy is to check the source, method and base of every figure before believing it.

## 7 Explain the limitations of statistics.

IMPORTANT

Statistics is a powerful tool, but it has clear limitations that every student must know.

First, statistics deals only with numerical facts, so qualities like honesty, beauty or intelligence cannot be studied directly unless converted into numbers.

Second, statistics studies groups and averages, never individuals. An average salary of Rs. 30,000 tells us nothing about what any single person earns.

Third, statistical results are true only on an average. They describe the general tendency of the group, not every single case, so they are never 100 percent exact.

Fourth, statistics can be deliberately misused. The same data can be presented in misleading ways, for example by choosing a convenient time period or by drawing a graph with an

exaggerated scale.

Fifth, statistics does not reveal the full story behind the numbers. Two classes can have the same average marks but very different spreads of marks.

Finally, statistical conclusions need expert interpretation. Wrong methods or careless reading lead to wrong decisions.

In short, statistics is a servant, not a master: it guides decisions but cannot replace judgement and common sense.

## 8 Distinguish between population and sample.

IMPORTANT

Population is the complete collection of all items or persons we want to study. A sample is only a part of the population that is actually observed.

For example, if we want to study the height of all BCA students in Kerala, all those students are the population, while 200 students selected from them are the sample.

A population is usually very large and hard to study fully, while a sample is small and manageable, so studying a sample saves time and money.

Information collected from a population is always complete and accurate, but it is expensive and slow. Information from a sample is quick and cheap, but it carries a small chance of error.

Results from a population are exact facts, while results from a sample are only estimates of the population, though a good sample gives a close estimate.

In short, the population is the whole field of study and the sample is a small representative part of it.

## 9 Define population, sample, parameter and statistic.

IMPORTANT

Population is the total set of all units under study. For example, all the students of a college form the population if we are studying that college.

Sample is a small part of the population selected for actual observation. For example, 50 students chosen from the college are a sample.

Parameter is a numerical value that describes the population. For example, the average height of ALL students of the college is a parameter. It is usually denoted by Greek letters like mu.

Statistic is a numerical value that describes the sample. For example, the average height of the 50 selected students is a statistic. It is usually denoted by English letters like x-bar.

A parameter is fixed but usually unknown, because studying the whole population is hard. A

statistic is known from the sample and is used to estimate the parameter.

In short, a parameter belongs to the population and a statistic belongs to the sample, and statistics uses the sample statistic to guess the population parameter.

## 10 Distinguish between primary and secondary data.

IMPORTANT

Primary data is data collected fresh for the first time for a specific purpose. Secondary data is data already collected by someone else for some other purpose and reused by us.

For example, interviewing 100 customers about a new product gives primary data, while using the government census report of 2011 gives secondary data.

Primary data is original and fits the present purpose exactly, but collecting it takes time and money. Secondary data is quick and cheap, but it may not fit our exact need.

Primary data is more reliable for our purpose because we control how it is collected. Secondary data may contain errors or outdated facts, so it must be checked before use.

Primary data is always collected by the investigator himself through surveys or experiments. Secondary data comes from published reports, newspapers, journals and websites.

In short, primary data is first-hand and tailor-made, while secondary data is second-hand and ready-made.

## 11 What are the methods of collecting primary data?

IMPORTANT

The main methods of collecting primary data are direct personal investigation, indirect oral investigation, questionnaires, schedules, and direct observation.

Direct personal investigation means the investigator himself meets the persons and collects answers. It gives accurate data, but it is costly and suitable only for a small area. For example, a researcher personally interviewing 30 shopkeepers in a town.

Indirect oral investigation means the investigator collects information from other persons who know about the subject, called witnesses. It is used when direct contact is not possible. For example, asking employers about workers' habits.

The questionnaire method means printed questions are sent by post or email to the informants, who fill and return them. It covers a wide area cheaply, but it works only with educated respondents.

The schedule method is similar, but the investigator himself visits the informant, asks the questions and fills the form. It gives better response than the postal questionnaire.

Direct observation means the investigator watches and records without asking questions. For example, counting the number of customers entering a shop per hour.

Telephone and online surveys are modern methods, quick and cheap, but they reach only people with phones or internet access.

## 12 What are the essentials of a good questionnaire?

IMPORTANT

A good questionnaire must have a clear purpose. Every question should be related to the aim of the study, and useless questions must be avoided.

The questions must be simple and easy to understand. Technical words and confusing language should not be used, because the respondent may answer wrongly.

Questions should be short and few in number. A long questionnaire bores the respondent and increases the chance that he will not reply.

Personal and embarrassing questions should be avoided, or asked very carefully, because the respondent may refuse to answer or give false answers.

Questions should be arranged in a logical order, starting with easy general questions and moving to specific ones, so the respondent feels comfortable.

Leading questions must be avoided. A question like 'Do you not agree our product is best?' pushes the answer, so neutral wording is needed.

Enough space must be given for answers, and instructions at the top must clearly explain how to fill the form.

A good questionnaire should be tested on a few people first, called a pilot survey, so that mistakes can be corrected before the real survey.

In short, a good questionnaire is simple, short, clear, unbiased and easy to fill.

## 13 Distinguish between qualitative and quantitative data with examples.

IMPORTANT

Qualitative data describes qualities or attributes that cannot be measured in numbers. Quantitative data consists of facts that can be measured and expressed in numbers.

Examples of qualitative data are gender, religion, colour, beauty, honesty and marital status. We can only classify them into groups, not measure them.

Examples of quantitative data are height, weight, age, income, marks and temperature. These can be counted or measured exactly.

Qualitative data is studied by counting how many units fall in each category, for example how

many students are male and how many are female.

Quantitative data is studied with averages and other numerical tools, for example the average income of workers or the average marks of a class.

Sometimes qualitative facts are converted into numbers, like rating beauty from 1 to 10, but the original fact is still a quality, not a true measurement.

In short, qualitative data tells us the kind of thing, while quantitative data tells us how much of it.

**14 Draft a questionnaire for collecting socio-demographic details of women workers in a factory. (2024)**

IMPORTANT

A questionnaire is a printed set of questions used to collect primary data directly from respondents.

For socio-demographic details of women workers, it must be simple, respectful and in the local language.

It should start with a short introduction stating the survey purpose and promising that answers stay confidential.

First come personal details: age group, marital status, number of children and education level.

Next come job details: department, designation, years of service, monthly income and shift timing.

Then come family details: type of house, total family income, distance from the factory and mode of transport.

Questions should be short and mostly tick-box type, so that even busy or less-educated workers can answer fast.

Avoid embarrassing or leading questions; never ask anything that hurts dignity or privacy.

Keep the form short, about 15 to 20 questions, because a long form tires workers and gives careless answers.

Example items: 'Age: below 25 / 25-35 / 35-45 / above 45' and 'Monthly income: below 10000 / 10000-20000 / above 20000'.

End with a thank-you line for the worker's time.

In short, this questionnaire is short, simple and respectful, covering personal, job and family details.

**15** **Explain the various types of sampling with examples.**

IMPORTANT

The main types of sampling are simple random sampling, stratified sampling, systematic sampling and cluster sampling, plus judgement sampling.

Simple random sampling means every unit has an equal chance of being selected. For example, writing the names of 500 students on slips, mixing them, and drawing 50 slips. It is the fairest method.

Stratified sampling means the population is first divided into groups called strata, like boys and girls, and then a random sample is taken from each group. It ensures every group is represented.

Systematic sampling means selecting every kth unit after a random start. For example, from a list of 1,000 customers, selecting every 10th customer. It is simple and quick.

Cluster sampling means the population is divided into clusters, like villages or classes, and a few whole clusters are selected at random. All units in the chosen clusters are studied.

Judgement sampling means the investigator selects the units using his own judgement, like choosing 20 shops he considers typical. It is quick but can be biased.

Probability methods like random, stratified, systematic and cluster sampling give every unit a known chance of selection, while judgement sampling is non-probability and depends on the investigator.

**16** **Distinguish between census and sample survey.**

IMPORTANT

A census survey studies every single unit of the population. A sample survey studies only a small selected part of it.

For example, the Indian census counts every person in the country, while an opinion poll interviews only a few thousand voters.

A census gives complete and exact information, but it takes a long time and costs a lot of money, so it is done rarely, like once in ten years.

A sample survey is quick and cheap, so it can be done often, but its results are only estimates and carry a small sampling error.

A census needs a huge staff and organisation. A sample survey needs fewer people and simpler organisation.

A census is suitable when the population is small or when full accuracy is needed. A sample survey is suitable when the population is large and quick results are needed.

In short, a census studies all and a sample studies a few, and the choice depends on time, money and the accuracy needed.

### 17 What is stratified sampling? Explain with an example.

IMPORTANT

Stratified sampling is a method in which the population is first divided into homogeneous groups called strata, and then a random sample is taken from each stratum.

The strata are formed on the basis of some common characteristic, like age, income, gender or class, so that units inside one stratum are similar.

Sampling from every stratum ensures that all types of units are represented in the sample, which a simple random sample may miss by chance.

For example, a college with 600 boys and 400 girls wants a sample of 100 students. It can take 60 boys at random from the boys' list and 40 girls at random from the girls' list.

The number taken from each stratum is usually in proportion to the size of the stratum, which is called proportionate stratified sampling.

This method gives more accurate results than simple random sampling when the population is mixed, because each group contributes its fair share to the sample.

### 18 Discuss on multi-stage sampling. (2024)

IMPORTANT

Multi-stage sampling is a probability sampling method in which the sample is selected in two or more stages.

At the first stage, large primary units are selected at random; at later stages, smaller units are picked from within them.

For example, to survey college students in Kerala: first select 5 districts at random, then 10 colleges from each district, then 20 students from each college.

It is very useful when the population is spread over a large area, because the investigator need not visit every corner.

It saves time and money compared to taking a simple random sample of the whole scattered population.

Its limitation is that sampling errors can add up at each stage, so the final sample may be less accurate.

In short, multi-stage sampling picks the sample step by step, from large groups down to the final units.

## 19 Discuss on systematic sampling. (2022)

IMPORTANT

Systematic sampling is a probability sampling method in which every kth unit of the population is selected after a random start.

First the population list is arranged in order; then the sampling interval is fixed as k = N/n, where N is population size and n is sample size.

A random number between 1 and k is chosen as the starting point, and then every kth unit is picked.

For example, to select 100 students from 1000: k = 1000/100 = 10; with a random start of 7, select roll numbers 7, 17, 27, 37 and so on.

Its merit is simplicity: only the first unit is chosen at random, the rest follow automatically, and the sample spreads evenly over the list.

Its limitation: if the list has a hidden repeating pattern that matches k, the sample becomes biased.

In short, systematic sampling selects every kth unit after one random start.

## 20 Explain the non-probability sampling techniques. (2022)

IMPORTANT

In non-probability sampling the units are not selected by chance; the investigator uses judgement or convenience.

The main types are judgement (purposive) sampling, convenience sampling, quota sampling and snowball sampling.

In judgement sampling the investigator deliberately picks units he considers typical, like selecting model farmers for an agriculture study.

In convenience sampling whoever is easily available is selected, like interviewing shoppers in one particular mall.

In quota sampling fixed quotas are set for each group, for example 50 men and 50 women, and the investigator fills them as he likes.

In snowball sampling one respondent leads to the next; it suits hidden populations whose members are hard to find directly.

The merit is speed and low cost; the limitation is that results cannot be safely generalised to the whole population.

In short, non-probability sampling is quick and cheap but not fully representative.

**21 What are the advantages of sample survey? (2018)**

IMPORTANT

A sample survey studies only a part of the population instead of every single unit.

Its first advantage is lower cost: studying 500 people costs far less than studying five lakh people.

Second, it saves time, so results come quickly, which matters for urgent decisions like election forecasts.

Third, the data is often more accurate, because a small team of investigators can be trained well and supervised closely.

Fourth, it is the only possible method when testing destroys the unit, like testing the life of electric bulbs.

Fifth, it needs fewer investigators and less organisation than a full census.

In short, a sample survey is cheaper, faster and often more reliable than a census.

**22 What are the principal steps involved in the planning and execution of a sample survey? (2018)**

IMPORTANT

A sample survey must be planned and executed in a clear sequence of steps.

First, define the objective: exactly what information is needed and for what purpose.

Second, define the population and prepare the sampling frame, the complete list of all units.

Third, choose the sampling method (simple random, stratified, systematic and so on) and fix the sample size.

Fourth, design the questionnaire or schedule and test it on a small pilot group.

Fifth, recruit and train the investigators, and fix the fieldwork timetable and budget.

Sixth, collect the data in the field with proper supervision and surprise checks.

Seventh, edit, classify and tabulate the collected data, removing errors and inconsistencies.

Finally, analyse the data, write the report and state the conclusions along with their limitations.

In short, a sample survey moves step by step from objective to frame to method to fieldwork to report.

## 23 What is tabulation? Explain the parts of a statistical table. (2024, 2018, 2019)

SURE MARK

Tabulation is the systematic arrangement of data in rows and columns in the form of a table, so that the data becomes easy to read and compare.

The first part of a table is the table number, which identifies the table, like Table 1 or Table 2.

The second part is the title, written at the top, which clearly states what the table is about, like 'Marks of BCA students, 2026'.

The third part is the stub, which is the left-hand column containing the row headings, like the names of classes or years.

The fourth part is the caption, which contains the column headings at the top, like 'Boys', 'Girls' and 'Total'.

The fifth part is the body of the table, which contains the actual figures arranged in rows and columns.

The sixth part is the head note, written just below the title in brackets, giving the unit of measurement, like '(in thousands)' or '(in Rs.)'.

The last part is the foot note, written at the bottom, which explains anything unusual about the data or its source.

A good table is simple, clear, well-titled and complete, so that anyone can understand the data at a glance.

## 24 What is classification? Explain its types.

IMPORTANT

Classification is the process of arranging data into groups or classes according to some common characteristic, so that similar facts come together.

The first type is geographical classification, where data is grouped by place. For example, population of India classified by states like Kerala, Tamil Nadu and Karnataka.

The second type is chronological classification, where data is grouped by time. For example, the sales of a company classified year-wise from 2020 to 2026.

The third type is qualitative classification, where data is grouped by attributes that cannot be measured. For example, students classified as male or female, or as pass or fail.

The fourth type is quantitative classification, where data is grouped by numerical values. For example, students grouped by marks like 0-10, 10-20 and 20-30.

Qualitative classification can be simple, with two classes like literate and illiterate, or manifold, with many classes like Hindu, Muslim and Christian.

Quantitative classification produces a frequency distribution, where each class shows how many units fall in it, like 25 students in the 60-70 marks class.

In short, classification organises raw data into meaningful groups, and it is the first step before tabulation.

**25** **Write a note on source note. (2024)**

IMPORTANT

A source note is a short line written below a statistical table stating where the data was taken from.

It is usually placed at the bottom left of the table, below the footnotes.

Its purpose is to give credit to the original source and to let the reader verify the figures if needed.

For example: 'Source: Census of India, 2011' or 'Source: Annual Report of XYZ Company, 2023-24'.

Without a source note the reader cannot judge whether the figures are trustworthy.

It is one of the essential parts of a good table, along with the title, captions, stubs, body and footnotes.

In short, the source note tells the reader the origin of the table's figures.

**26** **Form a frequency distribution from the following data by the exclusive method, taking 10 as the magnitude of class intervals: 10, 17, 15, 22, 11, 16, 19, 24, 29, 18, 25, 26, 32, 14, 17, 20, 23, 27, 30, 12. (2019)**

IMPORTANT

In the exclusive method the upper limit of a class is excluded and counted in the next class, so 10-20 means 10 up to but below 20.

First arrange the 20 values in order: 10, 11, 12, 14, 15, 16, 17, 17, 18, 19, 20, 22, 23, 24, 25, 26, 27, 29, 30, 32.

With class magnitude 10, the classes are 10-20, 20-30 and 30-40.

Class 10-20 takes 10, 11, 12, 14, 15, 16, 17, 17, 18, 19, giving frequency 10.

Class 20-30 takes 20, 22, 23, 24, 25, 26, 27, 29, giving frequency 8.

Class 30-40 takes 30, 32, giving frequency 2.

Check the total: 10 + 8 + 2 = 20, which equals the number of given values.

Answer: the frequency distribution is 10-20: 10, 20-30: 8, 30-40: 2.

**27** **Form a frequency distribution from the following data by the inclusive method, taking 4 as the magnitude of class intervals: 10, 17, 15, 22, 11, 16, 19, 24, 29, 18, 25, 26, 32, 14, 17, 20, 23, 27, 30, 12, 15, 18, 24, 36, 18, 15, 21, 28, 33, 38, 34, 13, 10, 16, 20, 22, 29, 19, 23, 31. (2022)**

IMPORTANT

In the inclusive method both limits of a class are included, so 10-13 means the values 10, 11, 12 and 13.

With magnitude 4 starting from 10, the classes are 10-13, 14-17, 18-21, 22-25, 26-29, 30-33, 34-37, 38-41.

Counting the 40 values into these classes: 10-13 gets 5 values (10, 10, 11, 12, 13).

14-17 gets 8 values (14, 15, 15, 15, 16, 16, 17, 17).

18-21 gets 8 values (18, 18, 18, 19, 19, 20, 20, 21).

22-25 gets 7 values (22, 22, 23, 23, 24, 24, 25).

26-29 gets 5 values (26, 27, 28, 29, 29).

30-33 gets 4 values (30, 31, 32, 33).

34-37 gets 2 values (34, 36) and 38-41 gets 1 value (38).

Check: 5+8+8+7+5+4+2+1 = 40, the total number of values.

Answer: frequencies are 5, 8, 8, 7, 5, 4, 2, 1 for the eight classes in order.

**28** **Write a short note on cross tabulation. (2022)**

IMPORTANT

Cross tabulation is a table that presents the joint distribution of two characteristics at the same time.

The rows show one characteristic and the columns show the other, and each cell gives the count for that combination.

For example, a table of 200 college students with rows Male/Female and columns Hosteller/Day scholar shows male hostellers, female hostellers and so on.

It is very useful for studying the relationship between two qualitative variables.

From a cross table we can compute row percentages and column percentages to compare groups easily.

Its limitation is that with more than two variables the table becomes large and difficult to read.

In short, cross tabulation classifies data by two characteristics together in a single table.

**29** **The following are the amounts of income tax paid by 600 businessmen in year 2010: Income tax Rs (more than) 500: 600, 1000: 550, 1500: 425, 2000: 275, 2500: 100, 3000: 25. Find the number of businessmen who paid more than Rs. 1,200 but not more than Rs. 2,400 as income tax. (2022)**

IMPORTANT

This is a 'more than' cumulative frequency table; first convert it into ordinary class frequencies.

500-1000: 600-550 = 50; 1000-1500: 550-425 = 125; 1500-2000: 425-275 = 150.

2000-2500: 275-100 = 175; 2500-3000: 100-25 = 75; above 3000: 25.

We need businessmen paying more than 1200 but not more than 2400.

From class 1000-1500 (125 persons), the part above 1200 is (1500-1200)/500 = 0.6, giving 0.6 x 125 = 75.

The full class 1500-2000 (150 persons) lies inside the range, so take all 150.

From class 2000-2500 (175 persons), the part up to 2400 is (2400-2000)/500 = 0.8, giving 0.8 x 175 = 140.

Total = 75 + 150 + 140 = 365.

Answer: 365 businessmen paid more than Rs. 1,200 but not more than Rs. 2,400.

**30** **How do you construct a pie diagram? Explain with the formula.**

IMPORTANT

A pie diagram is a circle divided into sectors, where each sector represents one part of the data and the size of the sector is proportional to the value it represents.

The first step is to find the total of all the values in the data.

The second step is to calculate the angle for each item using the formula: angle = (value of the item / total value) x 360 degrees, because a full circle has 360 degrees.

For example, if a family spends Rs. 900 on food, Rs. 600 on rent and Rs. 300 on others out of Rs. 1,800 total, the angles are: food = (900/1800) x 360 = 180 degrees, rent = (600/1800) x 360 = 120 degrees, others = (300/1800) x 360 = 60 degrees.

The third step is to draw a circle and divide it into sectors with these angles using a protractor, starting from a fixed point and moving clockwise.

Each sector is labelled or shaded differently, with a clear legend showing which sector stands for which item.

Pie diagrams are best when the data is divided into a few parts and we want to show the share of each part in the whole, like budget shares or market shares.

A pie diagram makes comparison of parts easy at a glance, because a bigger slice clearly

means a bigger share.

### 31 Write a short note on ogives.

IMPORTANT

An ogive is a graph of the cumulative frequency distribution, drawn as a smooth freehand curve. It is also called a cumulative frequency curve.

To draw it, the upper limits of the class intervals are marked on the x-axis and the cumulative frequencies on the y-axis, and the points are joined by a smooth curve.

There are two types of ogives. The 'less than' ogive is drawn using less-than cumulative frequencies, and its curve rises from left to right.

The 'more than' ogive is drawn using more-than cumulative frequencies, and its curve falls from left to right.

The main use of ogives is to find the median and other partition values like quartiles. The point where the two ogives cross gives the median of the data.

For example, to find the median marks of a class, we draw both ogives and read the x-value at their intersection point.

Ogives are also useful for comparing two distributions and for finding how many observations lie above or below a given value.

In short, an ogive turns a frequency table into a rising or falling curve that helps locate the median quickly.

### 32 Distinguish between a histogram and a bar diagram.

IMPORTANT

A histogram is drawn for continuous frequency data, while a bar diagram is drawn for discrete or categorical data.

In a histogram, the bars are joined together with no gaps, because the classes are continuous, like 0-10, 10-20. In a bar diagram, the bars are separated by gaps, because the categories are distinct, like years or products.

In a histogram, the x-axis shows class boundaries and the area of each bar represents the frequency. In a bar diagram, only the height of the bar represents the value.

A histogram can have unequal class widths, in which case the heights are adjusted so that the area stays proportional. A bar diagram always has bars of equal width.

A histogram cannot be drawn for qualitative data like gender or religion, but a bar diagram can easily show such categories.

For example, the marks of students in class intervals are shown by a histogram, while the sales of five different products are shown by a bar diagram.

In short, a histogram shows a continuous distribution with touching bars, while a bar diagram compares separate categories with spaced bars.

## 33 What is a frequency polygon? How is it drawn?

IMPORTANT

A frequency polygon is a line graph obtained by joining the mid-points of the tops of the rectangles of a histogram with straight lines.

To draw it, first the mid-points of each class are marked on the x-axis and the frequencies on the y-axis, and the points are plotted.

The plotted points are then joined by straight lines, forming a polygon, which gives the graph its name.

Two extra classes with zero frequency are added, one before the first class and one after the last class, so that the polygon touches the x-axis at both ends and looks complete.

A frequency polygon can also be drawn directly without first drawing a histogram, by plotting class mid-points against frequencies.

For example, for the marks classes 0-10, 10-20 and 20-30 with frequencies 5, 12 and 8, we plot the points (5, 5), (15, 12) and (25, 8) and join them.

Frequency polygons are useful for comparing two or more distributions on the same graph, because several polygons can be drawn together without confusion.

In short, a frequency polygon is a multi-sided line graph that shows the shape of a frequency distribution.

MODULE 2

# Measures of Central Tendency and Dispersion

2

**EXAM PATTERN (from real MGU question papers)**

EXAM PATTERN (QT): The real MG3CCRBCA200 paper pattern is UNKNOWN - no Honours model paper and no past BCA paper exists online (first exam 22-10-2025 was never published). Closest evidence: MGU B.Com 'Quantitative Techniques for Business' papers (same university, overlapping statistics syllabus) run 80 marks / 3 hours: Part A answer any 10 x 2 marks = 20 (short theory + one-line numericals), Part B answer any 6 x 5 marks = 30 (medium theory + numericals), Part C answer any 2 x 15 marks = 30 (long numericals: combined mean/SD/CV two-firm problem repeated 3x, Bayes theorem). QT is a 4-credit course, so the Honours paper may instead follow a 70-mark / 2-hour format (like DBMS) - this is NOT confirmed. Prepare every module: Part A 2-mark questions span all modules.

## 1 Define arithmetic mean. How is it calculated for individual, discrete and continuous series?

**IMPORTANT**

The arithmetic mean is the most common average. It is the sum of all the values divided by the number of values.

Formula: xbar = (sum of x) / N, where N is the total number of observations.

Individual series: simply add all the items and divide by N.

Example: mean of 4, 6, 8 = (4+6+8)/3 = 18/3 = 6.

Discrete series: when each value x has a frequency f, use xbar = (sum of f*x) / (sum of f).

Multiply each value by its frequency, add these products, and divide by the total frequency.

Continuous series: find the mid-point m of each class interval first.

Then use xbar = (sum of f*m) / (sum of f), where f is the class frequency.

The mid-point is calculated as (lower limit + upper limit) / 2.

Every value in the data is used in this average, so it represents the whole group.

In short, the formula changes only in what we multiply: single values, values times frequency, or mid-points times frequency.

**2** **Find the mean of 10, 12, 15, 18, 20.**

IMPORTANT

The formula for an individual series is xbar = (sum of x) / N.

Step 1: add all the values.

sum of x = 10 + 12 + 15 + 18 + 20.

= 22 + 15 + 18 + 20.

= 37 + 18 + 20.

= 55 + 20.

= 75.

Step 2: count the values. N = 5.

Step 3: divide the sum by N.

xbar = 75 / 5.

= 15.

Answer: mean = 15.

**3** **What are the merits and demerits of the arithmetic mean?**

IMPORTANT

The arithmetic mean is easy to calculate and simple to understand, even for a beginner.

It uses every value in the data, so no information is wasted.

It is rigidly defined, so two people always get the same answer from the same data.

It is useful for further mathematical work like combined means and standard deviation.

Merit summary: simple, stable, and mathematically the most workable average.

The main demerit is that extreme values pull it up or down and spoil it.

Example: mean of 2, 3, 4, 5, 86 is 20, which describes nobody in the group.

It cannot be found if the data is incomplete or has an open-ended class.

It cannot be used for qualitative data like favourite colour or honesty.

So the mean is best when the data is numerical and free from extreme values.

**4** **From the following data compute arithmetic mean by direct method: Marks 0-10: 5, 10-20: 10, 20-30: 25, 30-40: 30, 40-50: 20, 50-60: 10 students. (2019)**

IMPORTANT

For a continuous series the direct method uses class mid-points: Mean = sum(fx) / N.

Mid-points x are 5, 15, 25, 35, 45, 55 for the six classes.

Multiply each mid-point by its frequency: 5x5=25, 15x10=150, 25x25=625, 35x30=1050, 45x20=900, 55x10=550.

Total frequency N = 5+10+25+30+20+10 = 100.

Sum of fx = 25+150+625+1050+900+550 = 3300.

Mean = 3300 / 100 = 33.

Answer: the arithmetic mean is 33 marks.

**5** **The mean of 20 values is 45. If one of these values is to be taken 64 instead of 46, find the corrected mean. (2024)**

IMPORTANT

This is a corrected-mean problem: one wrong value must be replaced by the correct one.

Sum of the 20 values as first computed = 20 x 45 = 900.

Remove the wrong value 46 and add the correct value 64: corrected sum = 900 - 46 + 64 = 918.

The correct value 64 is larger than the wrong value 46, so the mean rises slightly after correction.

Corrected mean = 918 / 20 = 45.9.

Answer: the corrected mean is 45.9.

**6** **Compute median from the following data: Mid-values 115: 6, 125: 25, 135: 48, 145: 72, 155: 116, 165: 60, 175: 38, 185: 22, 195: 3. (2024, 2019)**

REPEATED

When mid-values are given, first reconstruct the classes; the gap between mid-values is 10, so classes are 110-120, 120-130, up to 190-200.

Total frequency N = 6+25+48+72+116+60+38+22+3 = 390, so N/2 = 195.

Cumulative frequencies: 6, 31, 79, 151, 267, 327, 365, 387, 390.

The median class is the first class whose cumulative frequency reaches 195, which is 140-150.

Here l = 140, previous cf = 151, f = 116, class width h = 10.

Median = l + ((N/2 - cf)/f) x h = 140 + ((195-151)/116) x 10 = 140 + 3.79 = 143.79.

Answer: the median is 143.79.

**7** **Define median. Find the median of 5, 8, 12, 15, 22.**

IMPORTANT

The median is the middle value of a dataset when all values are arranged in ascending order.

Half the observations lie below the median and half lie above it.

Odd N: median = the value at position (N+1)/2.

Even N: median = average of the values at positions N/2 and (N/2)+1.

Now find the median of 5, 8, 12, 15, 22.

The data is already arranged in order.

N = 5, which is odd.

Position of median = (5+1)/2 = 6/2 = 3rd value.

The 3rd value in 5, 8, 12, 15, 22 is 12.

Answer: median = 12.

**8** **Find the median of 4, 8, 10, 14.**

IMPORTANT

The data 4, 8, 10, 14 is already in ascending order.

N = 4, which is even, so the median is the average of the two middle values.

The two middle positions are N/2 = 2nd value and (N/2)+1 = 3rd value.

2nd value = 8 and 3rd value = 10.

Median = (8 + 10) / 2.

= 18 / 2.

= 9.

Answer: median = 9.

**9** **Why is the median preferred over the mean for income data?**

IMPORTANT

Income data is highly skewed because a few people earn very high incomes.

These few extreme incomes pull the arithmetic mean upward, far above what most people earn.

The median is not affected by extreme values because it depends only on the middle position.

So the median shows the income of the typical person much better than the mean.

Example: incomes 10, 12, 15, 18, 200 (in thousands) give mean 51 but median 15.

The median 15 clearly describes the common man, while the mean 51 misleads.

That is why the median is the preferred average for income and wealth data.

**10** **Define mode. Find the mode of 2, 3, 3, 5, 7, 3.**

IMPORTANT

The mode is the value that occurs most frequently in a dataset.

It is the value around which the maximum number of items cluster.

A dataset may have one mode, more than one mode, or no mode at all.

Now find the mode of 2, 3, 3, 5, 7, 3.

Count how many times each value appears.

2 appears once, 3 appears three times, 5 appears once, 7 appears once.

The value with the highest frequency is 3.

Answer: mode = 3.

**11** **When is the mode the best average to use?**

IMPORTANT

The mode is the best average when we want the most common or most popular value.

It is the only average that can be used for qualitative data.

Example: the most popular shirt size, the most liked colour, or the most common shoe size cannot be averaged by mean or median.

It is also useful when we need a quick, rough average without calculation.

Example: a shopkeeper wants to know which size sells most, so he uses the mode.

So use the mode whenever the question is about the most frequent item.

**12** **Calculate mode from the following: Marks (Above) 0: 122, 10: 118, 20: 112, 30: 92, 40: 60, 50: 27, 60: 10, 70: 2, 80: 0 students. (2024)**

IMPORTANT

This is a 'marks above' cumulative table; first convert it to ordinary frequencies by subtraction.

0-10: 122-118 = 4; 10-20: 118-112 = 6; 20-30: 112-92 = 20; 30-40: 92-60 = 32.

40-50: 60-27 = 33; 50-60: 27-10 = 17; 60-70: 10-2 = 8; 70-80: 2-0 = 2.

The modal class is 40-50 with the highest frequency 33.

Mode = l + ((f1-f0)/(2f1-f0-f2)) x h = 40 + ((33-32)/(66-32-17)) x 10.

This gives 40 + (1/17) x 10 = 40 + 0.59 = 40.59.

Answer: the mode is 40.59 marks.

**13** **State the empirical relationship between mean, median and mode.**

IMPORTANT

For a moderately skewed distribution, the three averages are connected by an empirical formula.

Formula: Mode = 3*Median - 2*Mean.

It is called empirical because it comes from observed data, not from a strict mathematical proof.

It works only for moderately skewed distributions.

If any two of the three are known, the third can be found using this relation.

Example: if mean = 20 and median = 18, then mode = 3*18 - 2*20.

= 54 - 40.

= 14.

So the mode is 14.

**14** **If mean = 20 and median = 18, find the mode.**

IMPORTANT

The empirical relationship is: Mode = 3*Median - 2*Mean.

Given: mean = 20 and median = 18.

Substitute the values.

Mode = 3*18 - 2*20.

Step 1: multiply. 3*18 = 54.

Step 2: multiply. 2*20 = 40.

Step 3: subtract. 54 - 40 = 14.

Answer: mode = 14.

15 **How do you locate the median graphically?**

IMPORTANT

Draw the less-than ogive (cumulative frequency plotted against upper class limits).

On the same graph, draw the more-than ogive (cumulative frequency against lower class limits).

Find the point where the two ogives intersect each other.

From that intersection point, drop a vertical line down to the x-axis.

The value on the x-axis where this line touches is the median.

This works because both ogives meet exactly at the N/2th value.

Graphical median is useful when the data is grouped in classes.

16 **How do you locate the mode graphically?**

IMPORTANT

First draw the histogram of the frequency distribution.

Find the tallest bar; it represents the modal class, where the mode lies.

Join the top-left corner of this tallest bar to the top-left corner of the next bar.

Join the top-right corner of the tallest bar to the top-right corner of the previous bar.

The two lines cross at a point above the modal class.

From this crossing point, drop a vertical line down to the x-axis.

The value where the line meets the x-axis is the mode.

This method is used when exact calculation is long.

17 **Find the weighted mean of prices 10, 20, 30 with weights 2, 3, 5. (2019, 2022)**

REPEATED

The formula for weighted mean is xbar_w = (sum of w*x) / (sum of w).

Given: prices x = 10, 20, 30 with weights w = 2, 3, 5.

Step 1: multiply each price by its weight.

10*2 = 20.

20*3 = 60.

30*5 = 150.

Step 2: add the products. 20 + 60 + 150 = 230.

Step 3: add the weights. 2 + 3 + 5 = 10.

Step 4: divide. 230 / 10 = 23.

Answer: weighted mean = 23.

**18** **The following data gives the weekly wages of workers in a firm, their total working hours and the average working hours per worker. Calculate the average weekly wage per worker. Wages Rs 80-100: total hours 168, avg hours 12; 100-120: 170, 10; 120-140: 225, 9; 140-160: 272, 8.5; 160-180: 126, 7; 180-200: 91, 6.5. (2019, 2022)**

REPEATED

This is a weighted-mean problem: first find the number of workers in each wage group.

Workers = total hours / average hours: 168/12 = 14, 170/10 = 17, 225/9 = 25, 272/8.5 = 32, 126/7 = 18, 91/6.5 = 14.

Total workers = 14+17+25+32+18+14 = 120.

Take wage mid-points 90, 110, 130, 150, 170, 190 as the values and worker counts as weights.

Weighted total = 90x14 + 110x17 + 130x25 + 150x32 + 170x18 + 190x14 = 1260+1870+3250+4800+3060+2660 = 16900.

Average weekly wage = 16900 / 120 = 140.83.

Answer: the average weekly wage per worker is Rs. 140.83.

**19** **Find the combined mean: group 1 has 30 students with mean 60, group 2 has 20 students with mean 70.**

IMPORTANT

The combined mean merges two groups into one single average, giving each group's mean a weight equal to its size.

Formula: xbar12 = (n1*xbar1 + n2*xbar2) / (n1 + n2), where n1 and n2 are the group sizes and xbar1, xbar2 are their means.

Here n1 = 30, xbar1 = 60 for group 1, and n2 = 20, xbar2 = 70 for group 2.

Substituting the values: xbar12 = (30*60 + 20*70) / (30 + 20).

First multiply: 30*60 = 1800 and 20*70 = 1400.

Then add: 1800 + 1400 = 3200, and total students = 30 + 20 = 50.

So xbar12 = 3200 / 50 = 64.

Answer: combined mean = 64.

The combined mean lies between the two group means, closer to 60 because the larger group of 30 students carries more weight.

**20** **Distinguish between combined mean and weighted mean.**

IMPORTANT

The combined mean joins the means of two or more groups into one overall mean.

The weighted mean joins values where each value is given an importance weight.

In the combined mean, the group sizes n1, n2 act as the weights automatically.

In the weighted mean, the weights are chosen by us to show importance.

Formula for combined mean: xbar12 = (n1*xbar1 + n2*xbar2) / (n1 + n2).

Formula for weighted mean: xbar_w = (sum of w*x) / (sum of w).

Example of combined mean: average marks of two classes of different sizes.

Example of weighted mean: exam subjects with different weightage, like 40% for internals and 60% for finals.

So the combined mean merges groups, while the weighted mean stresses important items.

**21** **The mean age of a group of 100 children was 9.35 years. The mean age of 25 of them was 8.75 years and that of another 65 was 10.51 years. What was the mean age for the remainder? (2024)**

IMPORTANT

This is a combined-mean problem solved backwards to find the missing group's mean.

Total age of all 100 children = 100 x 9.35 = 935 years.

Total age of the 25 children = 25 x 8.75 = 218.75 years.

Total age of the 65 children = 65 x 10.51 = 683.15 years.

Number in the remainder group = 100 - 25 - 65 = 10 children.

Their total age = 935 - 218.75 - 683.15 = 33.10 years.

Mean age of the remainder = 33.10 / 10 = 3.31 years.

Answer: the mean age of the remaining 10 children is 3.31 years.

## 22 Define quartiles, deciles and percentiles.

IMPORTANT

Quartiles divide the ordered data into 4 equal parts: Q1, Q2, Q3.

Q1 is the first quartile; 25% of values lie below it and 75% above it.

Q2 is the second quartile; it is exactly the median, with 50% on each side.

Q3 is the third quartile; 75% of values lie below it and 25% above it.

Deciles divide the ordered data into 10 equal parts, named D1 to D9.

Example: D5 is the 5th decile, which is also the median.

Percentiles divide the ordered data into 100 equal parts, named P1 to P99.

Example: P50 is the 50th percentile, which is also the median.

So Q2 = D5 = P50 = median, and all these values show how the data is spread.

## 23 Find Q1 and Q3 of 3, 7, 8, 12, 15, 18, 21.

IMPORTANT

The data 3, 7, 8, 12, 15, 18, 21 is already in ascending order.

N = 7.

Formula: Q1 is the value at position (N+1)/4.

Q1 position = (7+1)/4 = 8/4 = 2nd value.

The 2nd value is 7.

Formula: Q3 is the value at position 3*(N+1)/4.

Q3 position = 3*(7+1)/4 = 3*8/4 = 24/4 = 6th value.

The 6th value is 18.

Answer: Q1 = 7 and Q3 = 18.

## 24 Find the range and quartile deviation of 3, 7, 8, 12, 15, 18, 21. (2024, 2022, 2019)

REPEATED

Range = maximum value - minimum value.

Here, maximum = 21 and minimum = 3.

Range = 21 - 3.

= 18.

Answer: range = 18.

Quartile deviation formula: QD = (Q3 - Q1) / 2.

For this data, Q1 = 7 and Q3 = 18 (found from the quartile positions).

QD = (18 - 7) / 2.

= 11 / 2.

= 5.5.

Answer: quartile deviation = 5.5.

**25** **Define range, quartile deviation and mean deviation with formulas.**

IMPORTANT

The range is the simplest measure of dispersion.

Formula: Range = maximum value - minimum value.

It shows the total spread of the data in one figure.

The quartile deviation covers only the middle 50% of the data.

Formula: QD = (Q3 - Q1) / 2.

It is half of the inter-quartile range and ignores extreme values.

The mean deviation is the average of the absolute deviations from an average.

Formula: MD = (sum of |x - xbar|) / N, taken about the mean.

Absolute bars mean we ignore the sign and only take the size of each deviation.

Together, range gives total spread, QD gives middle spread, and MD gives average spread.

**26** **Find the mean deviation about the mean for 4, 6, 8, 10, 12.**

IMPORTANT

Mean deviation about the mean: MD = (sum of |x - xbar|) / N.

Step 1: find the mean.

xbar = (4 + 6 + 8 + 10 + 12) / 5.

= 40 / 5.

= 8.

Step 2: find the absolute deviation of each value from 8.

|4 - 8| = |-4| = 4.

|6 - 8| = |-2| = 2.

|8 - 8| = |0| = 0.

|10 - 8| = |2| = 2.

|12 - 8| = |4| = 4.

Step 3: add the absolute deviations.

4 + 2 + 0 + 2 + 4 = 12.

Step 4: divide by N = 5.

MD = 12 / 5.

= 2.4.

Answer: mean deviation about the mean = 2.4.

**27** **Calculate mean deviation and its co-efficient: Marks 0-10: 4, 10-20: 6, 20-30: 10, 30-40: 20, 40-50: 10, 50-60: 6, 60-70: 4 students. (2022)**

IMPORTANT

Mean deviation about the mean = sum(f x |x - mean|) / N; first find the mean using mid-points.

Mid-points are 5, 15, 25, 35, 45, 55, 65 with frequencies 4, 6, 10, 20, 10, 6, 4; N = 60.

fx values: 20, 90, 250, 700, 450, 330, 260; sum = 2100, so mean = 2100/60 = 35.

Deviations |x-35| are 30, 20, 10, 0, 10, 20, 30.

f|x-mean| values: 120, 120, 100, 0, 100, 120, 120; sum = 680.

Mean deviation = 680/60 = 11.33.

Co-efficient of mean deviation = MD / mean = 11.33/35 = 0.32.

Answer: mean deviation = 11.33 and its co-efficient = 0.32.

**28** **Calculate mean deviation from mean from the following data: Age 0-10: 6, 10-20: 5, 20-30: 8, 30-40: 15, 40-50: 7, 50-60: 6, 60-70: 3 persons. (2024)**

IMPORTANT

First find the mean with class mid-points 5, 15, 25, 35, 45, 55, 65 and frequencies 6, 5, 8, 15, 7, 6, 3; N = 50.

fx values: 30, 75, 200, 525, 315, 330, 195; sum = 1670, so mean = 1670/50 = 33.4.

Deviations |x-33.4| are 28.4, 18.4, 8.4, 1.6, 11.6, 21.6, 31.6.

f|x-mean| values: 170.4, 92.0, 67.2, 24.0, 81.2, 129.6, 94.8; sum = 659.2.

Mean deviation = 659.2/50 = 13.18.

Answer: the mean deviation from mean is 13.18 years.

**29** **An analysis of monthly wages paid to workers in two firms A and B gives: Firm A - 550 workers, average wage 50, SD 90; Firm B - 650 workers, average wage 45, SD 120. (1) Which firm pays larger total monthly wages? (2) Find the combined mean and combined SD. (3) In which firm is there greater variability? (2024, 2019, 2022)**

SURE MARK

This is the most repeated QT numerical: combined mean, combined SD and coefficient of variation together.

(1) Total wages: Firm A = 550 x 50 = 27,500; Firm B = 650 x 45 = 29,250. So Firm B pays the larger total monthly wage.

(2) Combined mean = (27500 + 29250) / 1200 = 56750/1200 = 47.29.

For combined SD use: variance = [n1(SD1^2 + d1^2) + n2(SD2^2 + d2^2)] / (n1+n2), where d1, d2 are deviations of group means from the combined mean.

d1 = 50 - 47.29 = 2.71 and d2 = 45 - 47.29 = -2.29.

Numerator = 550 x (8100 + 7.34) + 650 x (14400 + 5.24) = 4459039 + 9363409 = 13822448.

Combined variance = 13822448 / 1200 = 11518.71, so combined SD = sqrt(11518.71) = 107.33.

(3) CV of A = (90/50) x 100 = 180%; CV of B = (120/45) x 100 = 266.67%. Greater variability is in Firm B.

Answer: (1) Firm B; (2) combined mean 47.29, combined SD 107.33; (3) Firm B is more variable.

**30** **What is the coefficient of variation? If SD = 4 and mean = 20, find CV. (2024, 2018)**

REPEATED

The coefficient of variation is a relative measure of dispersion.

Definition: it expresses the standard deviation as a percentage of the mean.

Formula: CV = (SD / xbar) * 100.

It has no units, so it can compare two datasets with different units or means.

A lower CV means the data is more consistent; a higher CV means more variation.

Now find CV when SD = 4 and mean = 20.

CV = (4 / 20) * 100.

= 0.2 * 100.

= 20.

Answer: CV = 20%.

## 31 Define standard deviation and variance with formulas.

IMPORTANT

The standard deviation measures how far the values are spread from the mean.

Definition: SD is the square root of the average of the squared deviations from the mean.

Formula: SD = sqrt( sum((x - xbar)^2) / N ).

It is the best and most widely used measure of dispersion.

SD is measured in the same units as the original data.

The variance is simply the square of the standard deviation.

Formula: Variance = sum((x - xbar)^2) / N = (SD)^2.

Variance is measured in squared units, like square rupees.

So SD gives spread in original units, and variance gives spread in squared units.

## 32 Find the SD and variance of 2, 4, 6, 8, 10.

IMPORTANT

Step 1: find the mean.

xbar = (2 + 4 + 6 + 8 + 10) / 5.

= 30 / 5.

= 6.

Step 2: find the squared deviation of each value from 6.

(2 - 6)^2 = (-4)^2 = 16.

(4 - 6)^2 = (-2)^2 = 4.

(6 - 6)^2 = 0^2 = 0.

(8 - 6)^2 = 2^2 = 4.

(10 - 6)^2 = 4^2 = 16.

Step 3: add the squared deviations.

16 + 4 + 0 + 4 + 16 = 40.

Step 4: variance = sum of squared deviations / N = 40 / 5 = 8.

Step 5: SD = sqrt(variance) = sqrt(8).

sqrt(8) = 2.83 approx.

Answer: variance = 8 and SD = 2.83.

**33** **Two factories have CV 12% and 18%. Which is more consistent?**

IMPORTANT

The coefficient of variation measures consistency: lower CV means less variation and more consistency.

Factory 1 has CV = 12%, factory 2 has CV = 18%.

12% is smaller than 18%, so factory 1 has less relative variation.

Less variation means the performance of factory 1 is more stable and uniform.

Therefore the factory with CV 12% is more consistent.

Answer: the factory with CV 12% is more consistent.

**34** **If n = 10, sum x = 60, sum x-square = 1,000, calculate standard deviation. (2024)**

IMPORTANT

When only the sums are given, use the formula SD = sqrt[(sum x-square / n) - (sum x / n)^2].

sum x-square / n = 1000/10 = 100.

sum x / n = 60/10 = 6, and its square is 36.

Variance = 100 - 36 = 64.

SD = sqrt(64) = 8.

Answer: the standard deviation is 8.

MODULE 3

3

# Correlation and Regression

**EXAM PATTERN (from real MGU question papers)**

EXAM PATTERN (QT): The real MG3CCRBCA200 paper pattern is UNKNOWN - no Honours model paper and no past BCA paper exists online (first exam 22-10-2025 was never published). Closest evidence: MGU B.Com 'Quantitative Techniques for Business' papers (same university, overlapping statistics syllabus) run 80 marks / 3 hours: Part A answer any 10 x 2 marks = 20 (short theory + one-line numericals), Part B answer any 6 x 5 marks = 30 (medium theory + numericals), Part C answer any 2 x 15 marks = 30 (long numericals: combined mean/SD/CV two-firm problem repeated 3x, Bayes theorem). QT is a 4-credit course, so the Honours paper may instead follow a 70-mark / 2-hour format (like DBMS) - this is NOT confirmed. Prepare every module: Part A 2-mark questions span all modules.

## 1 Define correlation and explain its types with examples.

IMPORTANT

Correlation is a statistical measure that shows how strongly two variables move together.

It tells us whether a change in one variable is linked with a change in the other.

Positive correlation: both variables increase together. Example: height and weight of students.

Negative correlation: one variable increases while the other decreases. Example: price of a product and its demand.

Zero correlation: there is no relationship between the two variables. Example: shoe size and exam marks.

The value of correlation lies between -1 and +1.

Correlation is a very useful tool in business, economics and science to find hidden relationships.

## 2 Distinguish between linear and non-linear correlation.

IMPORTANT

Linear correlation exists when the two variables change in a constant ratio.

In linear correlation the points on a scatter diagram lie close to a straight line.

Example: if income rises by Rs. 1000 every year and savings rise by Rs. 200 every year, the relation is linear.

Non-linear correlation exists when the ratio of change between the two variables is not constant.

In non-linear correlation the points on a scatter diagram form a curve, not a straight line.

Example: at first a small rise in advertising gives a big rise in sales, but later extra advertising gives very little extra sales.

So the key difference is a straight-line pattern versus a curved pattern.

**3** **Explain the meaning and significance of the concept of correlation. Does it always imply cause and effect? (2024)**

IMPORTANT

Correlation measures the strength and direction of the linear relationship between two variables, with r lying between -1 and +1.

It is significant because it lets us predict one variable from another, like estimating sales from advertising spend.

It helps managers, scientists and economists discover hidden relationships in data before taking decisions.

However, correlation does NOT always imply cause and effect.

Two variables can move together because of a third common cause: ice-cream sales and drowning deaths both rise in summer, but one does not cause the other.

Such misleading cases are called spurious or nonsense correlation.

Therefore a high correlation is a clue, not a proof; causation needs controlled study.

In short, correlation shows association between variables, but cause and effect must be established separately.

**4** **What is a scatter diagram? How do you read correlation from it?**

IMPORTANT

A scatter diagram is a graph on which each pair of (x, y) values is plotted as a dot.

It is the simplest visual method to see the relationship between two variables.

To read correlation from it, look at the direction and closeness of the dots.

Dots rising from left to right show positive correlation.

Dots falling from left to right show negative correlation.

Dots scattered all over with no pattern show zero or no correlation.

Tightly packed dots in a narrow band mean strong correlation; widely spread dots mean weak correlation.

**5** **What are the merits and limitations of the scatter diagram?**

IMPORTANT

Merits:

A scatter diagram is very simple to draw and easy to understand.

It shows the pattern and direction of the relationship at a glance.

Odd or unusual values stand out clearly on the diagram.

It needs no complicated calculation.

Limitations:

It gives only a rough idea and no exact numerical measure of correlation.

Two people may judge the same diagram differently, so it is not fully objective.

It is not useful when the number of observations is very large.

Despite its limits, it is the best first step before calculating the correlation coefficient.

**6** **Calculate the coefficient of correlation for x: 1,2,3,4,5 and y: 2,3,5,4,6. (2024, 2022)**

SURE MARK

Formula: r = [N*sum(xy) - sum(x)*sum(y)] / sqrt([N*sum(x^2) - (sum(x))^2] * [N*sum(y^2) - (sum(y))^2])

Here N = 5.

sum(x) = 1+2+3+4+5 = 15.

sum(y) = 2+3+5+4+6 = 20.

sum(xy) = (1*2)+(2*3)+(3*5)+(4*4)+(5*6) = 2+6+15+16+30 = 69.

sum(x^2) = 1+4+9+16+25 = 55.

sum(y^2) = 4+9+25+16+36 = 90.

Numerator = N*sum(xy) - sum(x)*sum(y) = 5*69 - 15*20 = 345 - 300 = 45.

First term of denominator = N*sum(x^2) - (sum(x))^2 = 5*55 - 225 = 275 - 225 = 50.

Second term of denominator = N*sum(y^2) - (sum(y))^2 = 5*90 - 400 = 450 - 400 = 50.

Denominator = sqrt(50 * 50) = 50.

r = 45 / 50 = 0.9.

Answer: r = 0.9, which means strong positive correlation.

## 7 State the formula of Karl Pearson's coefficient of correlation and its properties.

IMPORTANT

Karl Pearson's coefficient of correlation is denoted by r and its formula is:

r = [N*sum(xy) - sum(x)*sum(y)] / sqrt([N*sum(x^2) - (sum(x))^2] * [N*sum(y^2) - (sum(y))^2])

Properties:

r always lies between -1 and +1.

r = +1 means perfect positive correlation and r = -1 means perfect negative correlation.

r = 0 means there is no linear relationship between the variables.

r measures only linear relation; it cannot catch a curved relationship.

It has no unit, because it is a pure number.

It needs both variables to be in numerical form.

It is affected by extreme values, so one odd value can change r a lot.

## 8 Explain the underlying assumptions of Karl Pearson's co-efficient of correlation. (2024)

IMPORTANT

Karl Pearson's coefficient r is valid only when certain assumptions about the data hold.

First, the relationship between the two variables must be linear; r cannot measure a curved relationship properly.

Second, both variables must be measured on an interval or ratio scale, that is, proper numerical data.

Third, the observations must be meaningfully paired, with a sensible link between each x and its y.

Fourth, the data should be free from extreme values, because a single outlier can distort r heavily.

Fifth, for full reliability the variables are assumed to follow roughly a normal distribution.

If these assumptions fail, rank correlation or other non-parametric methods are safer.

In short, Pearson's r assumes linear, numerical, outlier-free paired data.

**9** **If the covariance between X and Y is 488 and variance of X and Y are 824 and 325 respectively, find out coefficient of correlation. (2022)**

IMPORTANT

The shortcut formula is r = Cov(X,Y) / sqrt(VarX x VarY).

Here Cov = 488, VarX = 824 and VarY = 325.

Denominator = sqrt(824 x 325) = sqrt(267800) = 517.50.

r = 488 / 517.50 = 0.94.

The value is close to +1, showing a high degree of positive correlation between X and Y.

Answer: the coefficient of correlation is 0.94.

**10** **Compute Pearson's coefficient of correlation from the following data taking 60 and 70 as the assumed averages of X and Y respectively. Also see if it comes between 0.85 and 0.95. X: 45, 55, 56, 58, 60, 65, 68, 70, 75, 80, 85; Y: 56, 50, 48, 60, 62, 64, 65, 70, 74, 82, 90. (2022)**

IMPORTANT

With assumed means Ax = 60 and Ay = 70, take dx = X - 60 and dy = Y - 70 for the 11 pairs.

Sums: sum dx = 57, sum dy = -49, sum dx-square = 1709, sum dy-square = 1865, sum dx.dy = 1148.

r = [n.sum dx.dy - sum dx.sum dy] / sqrt[(n.sum dx-square - (sum dx)^2)(n.sum dy-square - (sum dy)^2)].

Numerator = 11x1148 - 57x(-49) = 12628 + 2793 = 15421.

Denominator = sqrt[(18799 - 3249)(20515 - 2401)] = sqrt(15550 x 18114) = 16783.2.

r = 15421 / 16783.2 = 0.92.

0.92 lies between 0.85 and 0.95, so the check is satisfied.

Answer: r = 0.92, which is between 0.85 and 0.95.

## 11 What is probable error? How do you interpret correlation using it?

IMPORTANT

Probable error (PE) is a measure used to test whether an observed correlation is real or just accidental.

Its formula is: PE = 0.6745 * (1 - r^2) / sqrt(N).

Here r is the correlation coefficient and N is the number of pairs.

Interpretation rules:

If r is greater than 6 times PE, the correlation is significant and definitely exists.

If r is less than PE, there is practically no correlation.

If r lies between PE and 6 times PE, we cannot say anything surely; the data may be small.

Probable error is useful because a high r from a small sample can still be doubtful.

## 12 If r = 0.9 and N = 25, find the probable error and comment.

IMPORTANT

Formula: PE = 0.6745 * (1 - r^2) / sqrt(N).

Given r = 0.9 and N = 25.

r^2 = 0.9 * 0.9 = 0.81.

1 - r^2 = 1 - 0.81 = 0.19.

sqrt(N) = sqrt(25) = 5.

PE = 0.6745 * 0.19 / 5.

First multiply: 0.6745 * 0.19 = 0.128155.

Then divide: 0.128155 / 5 = 0.025631.

So PE = 0.0256 approx.

6 * PE = 6 * 0.0256 = 0.1536.

Since r = 0.9 is greater than 6*PE = 0.1536, the correlation is highly significant.

Answer: PE = 0.0256; the correlation is significant.

**13** **What is the coefficient of determination? If r = 0.8, interpret it.**

IMPORTANT

The coefficient of determination is the square of the correlation coefficient, r^2.

It tells what fraction of the variation in one variable is explained by the other variable.

Its value lies between 0 and 1, and it is often written as a percentage.

For r = 0.8:

r^2 = 0.8 * 0.8 = 0.64.

This means 64 percent of the variation in y is explained by the variation in x.

The remaining 36 percent of the variation is due to other factors not included in the model.

So a higher r^2 means a better and more useful relationship for prediction.

**14** **State Spearman's rank correlation formula. When is it used? (2024, 2022)**

REPEATED

Spearman's rank correlation coefficient is denoted by rho and its formula is:

rho = 1 - [6 * sum(d^2)] / [n * (n^2 - 1)]

Here n is the number of pairs and d is the difference between the two ranks of each item.

When it is used:

It is used when data is given in the form of ranks, like 1st, 2nd, 3rd.

It is used for qualitative data that can be ranked, like beauty, honesty or taste.

It is also used when the data is not normally distributed or has extreme values.

Example: to find the relationship between the ranks of students in Maths and in Statistics.

**15** **Ranks in Maths: 1,2,3,4,5; ranks in Stats: 2,1,4,3,5. Find the rank correlation.**

IMPORTANT

Formula: rho = 1 - [6 * sum(d^2)] / [n * (n^2 - 1)]

Here n = 5.

Find d = Maths rank minus Stats rank for each student:

Student 1: d = 1 - 2 = -1, d^2 = 1.

Student 2: d = 2 - 1 = 1, d^2 = 1.

Student 3: d = 3 - 4 = -1, d^2 = 1.

Student 4: d = 4 - 3 = 1, d^2 = 1.

Student 5: d = 5 - 5 = 0, d^2 = 0.

sum(d^2) = 1 + 1 + 1 + 1 + 0 = 4.

Numerator = 6 * 4 = 24.

Denominator = 5 * (25 - 1) = 5 * 24 = 120.

rho = 1 - 24/120 = 1 - 0.2 = 0.8.

Answer: rho = 0.8, which means strong positive rank correlation.

**16 What is a tie in ranks? How is it corrected?**

IMPORTANT

A tie in ranks happens when two or more items have the same value, so they share the same rank.

Example: if two students tie for the 2nd and 3rd places, each is given the average rank 2.5.

The average rank given to the tied items keeps the total of all ranks correct.

Correction method:

For every group of ties, add (t^3 - t)/12 to sum(d^2).

Here t is the number of tied items in that group.

If there are two tie groups, apply this correction for each group separately.

The corrected sum(d^2) is then used in the formula rho = 1 - [6 * sum(d^2)] / [n * (n^2 - 1)].

**17 Write down any three merits and demerits of Spearman's rank correlation co-efficient. (2024)**

IMPORTANT

Merits: first, it is simple to understand and easy to calculate, simpler than Karl Pearson's method.

Second, it can be used even when the data is qualitative, like ranks given for beauty, honesty or intelligence.

Third, it is not affected much by extreme values, since only ranks matter, not actual magnitudes.

Demerits: first, it cannot be applied to grouped frequency distributions.

Second, it is less precise and less powerful than Pearson's coefficient for numerical data.

Third, when the number of observations is large, ranking all of them becomes tedious.

In short, rank correlation is simple and flexible, but it is less exact than Pearson's method.

## 18 Define regression. Explain its types.

IMPORTANT

Regression is a statistical method that studies the average relationship between variables for estimation and prediction.

While correlation only tells the strength of a relationship, regression gives an actual equation to predict one variable from another.

Types of regression:

Linear regression: the relationship is a straight line. Example: sales rise steadily with advertising.

Non-linear regression: the relationship is a curve. Example: production rises fast at first and then slows down.

Simple regression: there is one independent variable. Example: predicting marks (y) from study hours (x).

Multiple regression: there are two or more independent variables. Example: predicting marks (y) from study hours (x1) and attendance (x2).

Example: a shop uses regression to predict next month's sales (y) from this month's advertising spend (x).

## 19 What are regression lines? What is special about their intersection?

IMPORTANT

There are two regression lines:

The line of regression of y on x, which is used to predict y from a given x.

The line of regression of x on y, which is used to predict x from a given y.

Each line is the best fitting line for its own direction of prediction.

Special point about their intersection:

The two regression lines always cross each other at the point (xbar, ybar).

This point is the average of x and the average of y taken together.

If r = +1 or r = -1, that is perfect correlation, the two lines merge into a single line.

The smaller the angle between the two lines, the higher the correlation.

**20** **For x: 1,2,3,4,5 and y: 2,3,5,4,6, find the regression line of y on x. (2022)**

REPEATED

Formula for the regression line of y on x: y - ybar = byx * (x - xbar).

First find the means:

xbar = (1+2+3+4+5)/5 = 15/5 = 3.

ybar = (2+3+5+4+6)/5 = 20/5 = 4.

Now find byx = sum((x - xbar)(y - ybar)) / sum((x - xbar)^2).

Numerator terms (x - xbar)(y - ybar):

(1-3)(2-4) = (-2)(-2) = 4.

(2-3)(3-4) = (-1)(-1) = 1.

(3-3)(5-4) = 0*1 = 0.

(4-3)(4-4) = 1*0 = 0.

(5-3)(6-4) = 2*2 = 4.

Numerator = 4 + 1 + 0 + 0 + 4 = 9.

Denominator terms (x - xbar)^2:

(1-3)^2 = 4, (2-3)^2 = 1, (3-3)^2 = 0, (4-3)^2 = 1, (5-3)^2 = 4.

Denominator = 4 + 1 + 0 + 1 + 4 = 10.

byx = 9/10 = 0.9.

Line: y - 4 = 0.9 * (x - 3).

y = 0.9x - 2.7 + 4.

y = 0.9x + 1.3.

Answer: the regression line of y on x is y = 0.9x + 1.3.

**21** **Write the two regression equations and define the regression coefficients.**

IMPORTANT

The two regression equations are:

Regression of y on x: y - ybar = byx * (x - xbar).

Regression of x on y: x - xbar = bxy * (y - ybar).

Here xbar and ybar are the means of x and y.

Regression coefficients:

byx is the regression coefficient of y on x. Its formula is byx = r * (SDy / SDx).

It is the slope of the line of y on x and is used when predicting y.

bxy is the regression coefficient of x on y. Its formula is bxy = r * (SDx / SDy).

It is the slope of the line of x on y and is used when predicting x.

Both coefficients always have the same sign as the correlation coefficient r.

**22** **Write a note on least square method of computing regression equation. (2022)**

IMPORTANT

The least square method finds the regression line that fits the observed data best.

'Best' here means the line that makes the sum of the squared vertical distances of the points from the line as small as possible.

For y on x, we choose y = a + bx so that sum(y - a - bx)^2 is minimum.

Minimising gives the two normal equations: sum y = n.a + b.sum x and sum xy = a.sum x + b.sum x-square.

Solving these two equations gives the constants a and b, and hence the regression line.

The fitted line always passes through the point (mean of x, mean of y).

It is the most scientific method of fitting a regression line.

In short, least squares picks the line with the minimum total squared error.

**23** **What are the features of regression coefficients? (2022)**

IMPORTANT

The two regression coefficients byx and bxy have the following features.

First, both coefficients always carry the same sign, which is the sign of the correlation coefficient r.

Second, r is the geometric mean of the two coefficients: r = plus or minus sqrt(byx x bxy).

Third, if one regression coefficient is greater than 1, the other must be less than 1, because their product r-square cannot exceed 1.

Fourth, the coefficients are not symmetric: byx is generally not equal to bxy.

Fifth, both regression lines intersect at the point (x-bar, y-bar), the means of the two variables.

In short, the regression coefficients share the sign of r and are linked through r-square.

**24** **Given the following data, what would be the possible yield of rice per acre when rainfall is 29 cm? Coefficient of correlation between rainfall and yield is 0.8. Rainfall: Mean 25, Variance 9; Yield: Mean 40, Variance 36. (2022)**

IMPORTANT

Let x = rainfall and y = yield. SDx = sqrt(9) = 3 and SDy = sqrt(36) = 6.

Regression coefficient of y on x: byx = r x (SDy/SDx) = 0.8 x (6/3) = 1.6.

Since rainfall is the independent variable, use the regression line of y on x.

Regression line of y on x: y - y-bar = byx(x - x-bar), so y - 40 = 1.6(x - 25).

Put x = 29: y - 40 = 1.6 x 4 = 6.4, giving y = 46.4.

This estimate is reliable because the correlation 0.8 is fairly high.

Answer: the expected yield of rice is 46.4 per acre when rainfall is 29 cm.

**25** **Following data relates to marks in Accounts and Statistics: Accounts - Mean 30, SD 10; Statistics - Mean 35, SD 7; Coefficient of correlation 0.8. Find two regression equations and calculate the expected marks in Accounts if marks secured by a student in Statistics are 40. (2022)**

IMPORTANT

Let x = Accounts marks (mean 30, SD 10) and y = Statistics marks (mean 35, SD 7), with r = 0.8.

byx = r x (SDy/SDx) = 0.8 x (7/10) = 0.56; regression of y on x: y - 35 = 0.56(x - 30).

bxy = r x (SDx/SDy) = 0.8 x (10/7) = 1.143; regression of x on y: x - 30 = 1.143(y - 35).

To estimate Accounts (x) from Statistics (y = 40), use the x-on-y line.

x - 30 = 1.143 x (40 - 35) = 1.143 x 5 = 5.71, so x = 35.71.

Answer: the two regression equations are y - 35 = 0.56(x - 30) and x - 30 = 1.143(y - 35); expected Accounts marks = 35.71.

**26** **A panel of judges A and B graded seven debaters and independently awarded the following marks: A: 40, 34, 28, 30, 44, 38, 31; B: 32, 39, 26, 30, 38, 34, 28. An eighth debater was awarded 36 marks by Judge A while Judge B was not present. How many marks would you expect Judge B to award? (2022)**

IMPORTANT

This is a regression estimate: predict B's marks (y) from A's marks (x) using the y-on-x line.

Sums for n = 7: sum x = 245, sum y = 227, sum x-square = 8781, sum xy = 8066.

byx = [n.sum xy - sum x.sum y] / [n.sum x-square - (sum x)^2] = [56462 - 55615] / [61467 - 60025] = 847/1442 = 0.587.

Means: x-bar = 245/7 = 35, y-bar = 227/7 = 32.43.

Regression line: y - 32.43 = 0.587(x - 35).

For x = 36: y = 32.43 + 0.587 x 1 = 33.02.

Answer: Judge B would be expected to award about 33 marks.

**27** **Show that r^2 = byx * bxy.**

IMPORTANT

The two regression coefficients are defined as:

byx = r * (SDy / SDx).

bxy = r * (SDx / SDy).

Multiply the two coefficients:

byx * bxy = [r * (SDy / SDx)] * [r * (SDx / SDy)].

Group the terms: = r * r * (SDy / SDx) * (SDx / SDy).

The SD terms cancel: (SDy / SDx) * (SDx / SDy) = 1.

So byx * bxy = r^2.

This proves that the product of the two regression coefficients equals the square of the correlation coefficient.

Taking the square root: r = +sqrt(byx * bxy) or -sqrt(byx * bxy), with the sign of the coefficients.

**28** **If byx = 0.9 and bxy = 0.9, find r.**

IMPORTANT

Formula: r^2 = byx * bxy.

Substitute the values: r^2 = 0.9 * 0.9 = 0.81.

Take the square root: r = +sqrt(0.81) or -sqrt(0.81).

sqrt(0.81) = 0.9, so r = +0.9 or -0.9.

Both regression coefficients are positive, so r must also be positive.

Answer: r = 0.9.

**29** **Given 2x - y = 5 and 3x - 2y = 4, identify the regression lines and find r.**

IMPORTANT

We must decide which line is y on x and which is x on y, because r^2 = byx * bxy must be between 0 and 1.

First try: treat 2x - y = 5 as the line of y on x.

Rewrite as y = 2x - 5, so byx = 2.

Treat 3x - 2y = 4 as the line of x on y: 3x = 2y + 4, so x = (2/3)y + 4/3, giving bxy = 2/3.

Then r^2 = 2 * (2/3) = 4/3 = 1.33.

This is greater than 1, which is impossible, so this assignment is wrong.

So swap the roles: treat 2x - y = 5 as the line of x on y.

2x = y + 5, so x = 0.5y + 2.5, giving bxy = 0.5.

Treat 3x - 2y = 4 as the line of y on x.

2y = 3x - 4, so y = 1.5x - 2, giving byx = 1.5.

Now r^2 = byx * bxy = 1.5 * 0.5 = 0.75.

r = +sqrt(0.75) = 0.866 approx, positive because both coefficients are positive.

Answer: 2x - y = 5 is the line of x on y, 3x - 2y = 4 is the line of y on x, and r = 0.866.

## 30 Distinguish between correlation and regression.

IMPORTANT

Correlation measures the strength and direction of the relationship between two variables; regression gives the exact equation to predict one from the other.

Correlation is symmetric: the r between x and y is the same as between y and x. Regression is not symmetric: the line of y on x is different from the line of x on y.

Correlation lies between -1 and +1; regression coefficients can take any value and may exceed 1.

Correlation does not say which variable is cause and which is effect; regression treats x as the fixed independent variable and y as the dependent variable.

Correlation is not used for actual prediction, while regression is mainly used for estimation and forecasting.

Example: r = 0.9 only says marks and study hours move together strongly, while the regression equation lets us predict marks for 5 study hours.

MODULE 4

4

# Basic Concepts of Probability

**EXAM PATTERN (from real MGU question papers)**

EXAM PATTERN (QT): The real MG3CCRBCA200 paper pattern is UNKNOWN - no Honours model paper and no past BCA paper exists online (first exam 22-10-2025 was never published). Closest evidence: MGU B.Com 'Quantitative Techniques for Business' papers (same university, overlapping statistics syllabus) run 80 marks / 3 hours: Part A answer any 10 x 2 marks = 20 (short theory + one-line numericals), Part B answer any 6 x 5 marks = 30 (medium theory + numericals), Part C answer any 2 x 15 marks = 30 (long numericals: combined mean/SD/CV two-firm problem repeated 3x, Bayes theorem). QT is a 4-credit course, so the Honours paper may instead follow a 70-mark / 2-hour format (like DBMS) - this is NOT confirmed. Prepare every module: Part A 2-mark questions span all modules.

## 1 Define random experiment, sample space and event with examples.

**IMPORTANT**

A random experiment is an experiment whose outcome cannot be predicted in advance, but the set of all possible outcomes is known.

Example: tossing a coin is a random experiment because we cannot say whether Head or Tail will come, but we know the only possible outcomes are Head and Tail.

Sample space is the set of all possible outcomes of a random experiment. It is denoted by S.

Example: for tossing a coin, S = {H, T}. For throwing a die once, S = {1, 2, 3, 4, 5, 6}.

An event is a subset of the sample space. It is a collection of one or more possible outcomes that we are interested in.

Example: on a die, the event 'getting an even number' is {2, 4, 6}.

A simple event has only one outcome, like {5}; a compound event has more than one outcome, like {2, 4, 6}.

The sure event is the whole sample space S, and the impossible event is the empty set, which can never happen.

In short, the random experiment is the act, the sample space lists everything that can happen, and the event is the part of it we are interested in.

## 2 Explain mutually exclusive and exhaustive events with examples.

IMPORTANT

Mutually exclusive events are events that cannot happen at the same time in one trial of the experiment.

In set language, A and B are mutually exclusive if A intersect B is the empty set, meaning they have no common outcome.

Example: in one toss of a coin, getting Head and getting Tail are mutually exclusive, because both cannot appear in the same toss.

Exhaustive events are events that together cover all the possible outcomes of the sample space.

In set language, A and B are exhaustive if A union B = S, the whole sample space, so no outcome is left out.

Example: for a die, 'getting an even number' {2, 4, 6} and 'getting an odd number' {1, 3, 5} are exhaustive because together they cover all six outcomes.

These two events are also mutually exclusive, since no number is both even and odd at the same time.

So mutually exclusive means 'no common outcome', and exhaustive means 'no outcome left out'.

Two events can be both mutually exclusive and exhaustive, like Head and Tail, or neither, like {1, 2} and {2, 3} on a die.

## 3 List the sample space in throwing 2 dice. (2022)

IMPORTANT

When two dice are thrown together, each die shows 1 to 6, giving 6 x 6 = 36 equally likely outcomes.

The sample space S is the set of all ordered pairs (first die, second die).

It is: (1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6),

(3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6),

(5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6).

Every probability problem on two dice starts from this 36-point sample space.

Answer: the sample space has 36 ordered pairs as listed above.

**4** **Explain union, intersection and complement of events.**

IMPORTANT

Union of events: A union B is the event that happens when at least one of A or B happens, that is A or B or both.

Example: on a die, let A = {2, 4, 6} (even number) and B = {4, 5, 6} (number greater than 3). Then A union B = {2, 4, 5, 6}.

Intersection of events: A intersect B is the event that happens only when both A and B happen together.

Example: for the same A and B, A intersect B = {4, 6}, which are the outcomes common to both events.

Complement of an event: A complement is the event that happens when A does not happen. It contains all outcomes of the sample space that are not in A.

Example: if A = {2, 4, 6}, its complement is {1, 3, 5}.

In probability: P(A union B) = P(A) + P(B) - P(A intersect B), and P(A complement) = 1 - P(A).

Union, intersection and complement let us build complicated events from simple ones and compute their probabilities.

**5** **Represent the following by means of a Venn Diagram: A union B, A intersection B and A-complement. (2024)**

IMPORTANT

A Venn diagram draws events as circles inside a rectangle that stands for the sample space.

A union B is shown by shading the whole area covered by both circles, every outcome in A or B or both.

A intersection B is shown by shading only the overlapping lens where the two circles cross, outcomes common to both.

A-complement (A-bar) is shown by shading everything inside the rectangle except the A circle.

These three shadings are the visual form of the addition rule: P(A union B) = P(A) + P(B) - P(A intersection B).

In short, union shades both circles fully, intersection shades only the overlap, and complement shades outside A.

**6** **Define probability. Explain the classical, empirical and axiomatic definitions.**

IMPORTANT

Probability is a number between 0 and 1 that measures how likely an event is to happen.

A probability of 0 means the event is impossible; a probability of 1 means the event is certain.

Classical definition: P(A) = number of favourable outcomes / total number of equally likely outcomes.

It works when every outcome has the same chance, like a fair coin or a fair die.

Example: P(getting a 6 on a die) = 1/6.

Limitation: it cannot be used when outcomes are not equally likely, like a biased coin.

Empirical definition: P(A) = number of times A occurred / total number of trials, based on actual experiments and observation.

Example: if a coin tossed 1000 times gives 480 heads, the empirical probability of a head is 480/1000 = 0.48.

Limitation: the answer changes with the number of trials, so it is only approximate.

Axiomatic definition (given by Kolmogorov): probability is defined by three rules - (1) P(A) >= 0 for every event, (2) P(S) = 1 for the whole sample space, (3) if A and B are mutually exclusive then P(A union B) = P(A) + P(B).

This definition accepts the classical and empirical ideas as special cases and is the basis of all advanced probability.

In short: classical counts the outcomes, empirical observes the trials, and axiomatic lays down the rules.

**7** **A die is tossed once. Find P(getting a number greater than 4).**

IMPORTANT

Formula: P(A) = number of favourable outcomes / total number of equally likely outcomes.

The sample space for tossing a die once is S = {1, 2, 3, 4, 5, 6}.

Total number of equally likely outcomes = 6.

The numbers greater than 4 are 5 and 6.

Number of favourable outcomes = 2.

Substitute the values: P(getting a number greater than 4) = 2/6.

Simplify the fraction: 2/6 = 1/3.

Answer: P = 1/3.

**8** **What is the probability that a card drawn from a pack of 52 cards is a card of King? (2024)**

IMPORTANT

Use the classical definition: P(event) = favourable outcomes / total equally likely outcomes.

Total outcomes when one card is drawn = 52.

Favourable outcomes = 4 kings (one each of spades, hearts, diamonds, clubs).

P(King) = 4/52 = 1/13.

This is the simplest use of the classical definition: count the favourable cases over all equally likely cases.

Answer: the probability is 1/13.

**9** **A class consists of 4 girls and 3 boys to be arranged for a photograph in a single row. In how many ways can they be seated if all the girls sit together? (2024)**

IMPORTANT

Treat the 4 girls as one single block, since they must stay together.

Now we arrange 5 units: the girls-block plus the 3 boys, which can be done in 5! = 120 ways.

Inside the block, the 4 girls can arrange themselves in 4! = 24 ways.

By the multiplication rule, total arrangements = 120 x 24 = 2880.

The block trick works for any 'stay together' seating problem: bind the group, arrange the units, then arrange inside.

Answer: they can be seated in 2880 ways with all girls together.

**10** **In how many ways can a cricket team be selected from 25 players consisting of 10 batsmen, 8 bowlers, 5 all-rounders and 2 wicketkeepers, if the team requires 3 batsmen, 3 all-rounders, 2 bowlers and 1 wicketkeeper? (2022)**

IMPORTANT

This is a combination problem: choose the required players from each category and multiply.

3 batsmen from 10: C(10,3) = 120 ways.

3 all-rounders from 5: C(5,3) = 10 ways.

2 bowlers from 8: C(8,2) = 28 ways.

1 wicketkeeper from 2: C(2,1) = 2 ways.

Total selections = 120 x 10 x 28 x 2 = 67,200.

Answer: 67,200 ways for the specified 9 positions.

**11** **State and prove the addition theorem for two events.**

IMPORTANT

Statement: For any two events A and B, P(A union B) = P(A) + P(B) - P(A intersect B).

For mutually exclusive events, where A intersect B is empty, the formula simplifies to P(A union B) = P(A) + P(B).

Proof: Suppose the experiment has n(S) equally likely outcomes in the sample space S.

Let n(A) be the number of outcomes in event A, and n(B) the number of outcomes in event B.

Case 1 - mutually exclusive events: A and B have no common outcome, so the outcomes in A union B are simply n(A) + n(B).

That is, n(A union B) = n(A) + n(B).

Divide both sides by n(S): n(A union B)/n(S) = n(A)/n(S) + n(B)/n(S).

By the classical definition of probability, this gives P(A union B) = P(A) + P(B).

Case 2 - overlapping events: the common outcomes counted in n(A intersect B) are counted twice in n(A) + n(B), once in each event.

So we subtract them once: n(A union B) = n(A) + n(B) - n(A intersect B).

Dividing both sides by n(S) gives P(A union B) = P(A) + P(B) - P(A intersect B).

This completes the proof of the addition theorem for two events.

**12** **If P(A) = 0.4, P(B) = 0.5 and P(A intersect B) = 0.2, find P(A union B).**

IMPORTANT

Formula: P(A union B) = P(A) + P(B) - P(A intersect B).

Given: P(A) = 0.4, P(B) = 0.5, P(A intersect B) = 0.2.

Substitute the values: P(A union B) = 0.4 + 0.5 - 0.2.

First add the first two terms: 0.4 + 0.5 = 0.9.

Then subtract the overlap: 0.9 - 0.2 = 0.7.

Answer: P(A union B) = 0.7.

**13** **State the addition theorem for three events.**

IMPORTANT

Statement: For any three events A, B and C, P(A union B union C) = P(A) + P(B) + P(C) - P(A intersect B) - P(B intersect C) - P(A intersect C) + P(A intersect B intersect C).

Explanation: first add the probabilities of the three single events.

Then subtract the three pairwise intersections, because every common outcome is counted twice when the singles are added.

Finally add back the triple intersection, because those outcomes were subtracted too many times in the previous step.

For three mutually exclusive events all the overlap terms are zero, so the formula simplifies to P(A union B union C) = P(A) + P(B) + P(C).

Example: drawing one card from a pack, let A = king, B = queen, C = jack. Then P = 4/52 + 4/52 + 4/52 = 12/52 = 3/13.

This theorem extends the two-event addition rule by carefully adjusting for every overlap.

**14** **One number is drawn from numbers 1 to 100. Find the probability that it is either divisible by 3 or 5. (2022)**

IMPORTANT

This is a direct application of the addition theorem: P(A union B) = P(A) + P(B) - P(A intersection B).

Numbers from 1 to 100 divisible by 3: 33 numbers, so P(divisible by 3) = 33/100.

Numbers divisible by 5: 20 numbers, so P(divisible by 5) = 20/100.

Numbers divisible by both 3 and 5, that is by 15: 6 numbers, so P(both) = 6/100.

P(either) = 33/100 + 20/100 - 6/100 = 47/100.

Answer: the probability is 47/100 = 0.47.

**15** **Define conditional probability. A bag has 3 red and 2 blue balls; two are drawn without replacement. Find P(both red).**

IMPORTANT

Definition: Conditional probability is the probability of an event A given that another event B has already happened. It is written P(A|B).

Formula: P(A|B) = P(A intersect B) / P(B), where P(B) is greater than zero.

It tells us the chance of A after B has reduced the sample space.

The bag has 3 red and 2 blue balls, so there are 5 balls in total.

Formula for successive draws: P(both red) = P(first red) * P(second red | first red).

P(first red) = 3/5, because 3 of the 5 balls are red.

After one red ball is drawn without replacement, 2 red balls and 4 balls in total remain.

So P(second red | first red) = 2/4.

Substitute the values: P(both red) = (3/5) * (2/4).

Multiply the fractions: (3/5) * (2/4) = 6/20.

Simplify: 6/20 = 3/10 = 0.3.

Answer: P(both red) = 0.3.

**16** **Define independent events. Give an example.**

IMPORTANT

Definition: Two events A and B are independent if the happening of one does not affect the probability of the other.

In symbols, A and B are independent if P(A|B) = P(A), which is the same as P(A intersect B) = P(A) * P(B).

Knowing that B happened gives no information about whether A will happen.

Example: toss a coin twice. Let A = 'head on the first toss' and B = 'head on the second toss'.

The outcome of the first toss does not change the second toss, so P(B|A) = 0.5 = P(B).

Hence A and B are independent, and P(A intersect B) = 0.5 * 0.5 = 0.25.

In contrast, drawing two balls without replacement gives dependent events, because the first draw changes the chances for the second draw.

So independence means the two events have no influence on each other.

**17** **A bag contains 7 red, 12 white and 4 green balls. What is the probability that: (a) 3 balls drawn are all white and (b) 3 balls drawn are one of each colour? (2022)**

IMPORTANT

Total balls = 7 + 12 + 4 = 23; the number of ways to draw 3 balls is C(23,3) = 1771.

(a) Ways to draw 3 white balls from 12: C(12,3) = 220. So P(all white) = 220/1771.

(b) One of each colour: choose 1 red AND 1 white AND 1 green = 7 x 12 x 4 = 336 ways by the multiplication rule.

So P(one of each colour) = 336/1771.

Both parts use the classical definition with combinations, since the balls of the same colour are identical for counting.

Answer: (a) 220/1771 and (b) 336/1771.

**18** **A candidate is selected for interview for three posts. For the first post there are three candidates, for the second there are four and for the third there are two. What are the chances of his getting at least one post? (2024)**

IMPORTANT

For 'at least one' problems, use the complement: P(at least one) = 1 - P(getting none).

P(getting post 1) = 1/3, so P(missing it) = 2/3.

P(getting post 2) = 1/4, so P(missing it) = 3/4.

P(getting post 3) = 1/2, so P(missing it) = 1/2.

Assuming the selections are independent, P(missing all three) = (2/3) x (3/4) x (1/2) = 6/24 = 1/4.

P(at least one post) = 1 - 1/4 = 3/4.

Answer: the chance of getting at least one post is 3/4 = 0.75.

**19** **State the multiplication theorem for two and three events.**

IMPORTANT

For two events: P(A intersect B) = P(A) * P(B|A).

This follows from the conditional probability formula P(B|A) = P(A intersect B) / P(A), by multiplying both sides by P(A).

It can also be written the other way round: P(A intersect B) = P(B) * P(A|B).

For three events: P(A intersect B intersect C) = P(A) * P(B|A) * P(C | A intersect B).

The idea is the same: the first event happens, then the second given the first, then the third

given the first two together.

For independent events the conditions drop out, so for two independent events P(A intersect B) = P(A) * P(B).

For three independent events: P(A intersect B intersect C) = P(A) * P(B) * P(C).

Example of the independent case: P(two heads in two tosses) = 0.5 * 0.5 = 0.25.

**20** **A coin is tossed three times. Find P(getting three heads).**

IMPORTANT

The three tosses are independent, and for each toss P(head) = 1/2.

Formula for three independent events: P(A intersect B intersect C) = P(A) * P(B) * P(C).

Substitute the values: P(three heads) = (1/2) * (1/2) * (1/2).

Multiply the first two factors: (1/2) * (1/2) = 1/4.

Multiply by the third factor: (1/4) * (1/2) = 1/8.

Convert to decimal: 1/8 = 0.125.

Answer: P(getting three heads) = 1/8 = 0.125.

**21** **A University has to select an examiner from 50 persons: 20 women and 30 men; 10 knowing Hindi and 40 not; 15 teachers and 35 not. What is the probability of selecting a Hindi knowing woman teacher? (2022)**

IMPORTANT

This applies the multiplication theorem for three independent attributes.

P(woman) = 20/50 = 0.4; P(knowing Hindi) = 10/50 = 0.2; P(teacher) = 15/50 = 0.3.

Treating the three attributes as independent, multiply the probabilities.

P(Hindi-knowing woman teacher) = 0.4 x 0.2 x 0.3 = 0.024.

This works because the three attributes are treated as independent choices from the list of 50 persons.

Answer: the probability is 0.024, that is 2.4%.

**22** **A factory has Machine A making 60% of items (2% defective) and Machine B making 40% (5% defective). An item is defective; find P(it came from A). (Apr 2022, May 2022)**

REPEATED

Let A = item came from Machine A, B = item came from Machine B, D = item is defective.

Given: P(A) = 0.6, P(D|A) = 0.02; P(B) = 0.4, P(D|B) = 0.05.

Formula (Bayes): P(A|D) = [P(A) * P(D|A)] / [P(A)*P(D|A) + P(B)*P(D|B)].

Numerator: P(A) * P(D|A) = 0.6 * 0.02.

Multiply: 0.6 * 0.02 = 0.012.

Denominator second term: P(B) * P(D|B) = 0.4 * 0.05.

Multiply: 0.4 * 0.05 = 0.020.

Total probability of defective: P(D) = 0.012 + 0.020 = 0.032.

Substitute into the formula: P(A|D) = 0.012 / 0.032.

Divide: 0.012 / 0.032 = 0.375.

Answer: P(it came from Machine A) = 0.375.

**23** **State Bayes theorem.**

IMPORTANT

Bayes theorem reverses the condition: it finds P(Ai|B) from P(B|Ai).

Statement: If A1, A2, ..., An are mutually exclusive and exhaustive events (the possible causes) and B is any event, then P(Ai|B) = [P(Ai) * P(B|Ai)] / [P(A1)*P(B|A1) + P(A2)*P(B|A2) + ... + P(An)*P(B|An)].

P(Ai) is the prior probability of cause Ai, and P(B|Ai) is the probability of the effect B if cause Ai is true.

The numerator is the chance that cause Ai produced the observed effect B.

The denominator is the total chance of B happening through any of the causes, found by the total probability rule.

It is used when we see the effect (B has happened) and want to find the most probable cause.

For two causes it simplifies to P(A1|B) = [P(A1)*P(B|A1)] / [P(A1)*P(B|A1) + P(A2)*P(B|A2)].

In short, Bayes theorem turns 'probability of effect given cause' into 'probability of cause given effect'.

**24** **Give two applications of Bayes theorem.**

IMPORTANT

Application 1: Medical diagnosis. A doctor knows the accuracy of a blood test and uses Bayes theorem to find the probability that a patient actually has the disease given that the test result is positive.

The prior is how common the disease is in the population; the positive test result updates it to a posterior probability for that particular patient.

Application 2: Spam filtering. An email program uses Bayes theorem to find the probability that a mail is spam given that certain words appear in it.

From past data it learns how often each word appears in spam mails versus normal mails, and updates the probability as more words are found.

A third example is quality control in factories, where a defective item found on the line is traced back to the machine that most likely made it.

In all these cases Bayes theorem turns an observed effect into the most probable cause.

**25** **Your officer is happy on 60% of your calls, so P(happy on visit) = 0.6. If he is happy he accedes to your request with probability 0.4, and if not happy with probability 0.1. You call one day and he accedes. What is the probability he was happy? (2022)**

IMPORTANT

This is a Bayes theorem problem: turn the observed effect (he acceded) back into the probable cause (he was happy).

P(Happy) = 0.6 and P(Not happy) = 0.4.

P(Accede | Happy) = 0.4 and P(Accede | Not happy) = 0.1.

Total P(Accede) = 0.6x0.4 + 0.4x0.1 = 0.24 + 0.04 = 0.28.

By Bayes: P(Happy | Accede) = (0.6 x 0.4) / 0.28 = 0.24/0.28 = 0.857.

Answer: the probability he was happy is 0.86 (exactly 6/7).